

Front End vs Back End

1. What is Front End?

- Front End is the part of a system where users interact with the data and create reports, dashboards and visualizations.
- In Power BI, it includes the Report View, Visualizations, Filters, Slicers, and the user interface (UI).
- It focuses on how the data is presented to the end user.

Examples:

- Reports, charts, graphs, dashboards
- Buttons, slicers, filters
- Formatting and design



Front End

(What users see and interact with)

2. What is Back End?

- Back End is the part where data is collected, transformed, and prepared before it is shown in the front end.
- In Power BI, it includes Data Sources, Power Query Editor (data transformation), Data Model and relationships.
- It handles the data logic and structure.

Examples:

- Data source connections
- Power Query (ETL/Transformations)
- Data Model (tables, relationships, DAX)



Back End

(Where data is prepared and structured)

3. Data Engineer vs Data Analyst - How They Work in a Company

Data Engineer

- Builds and maintains the data pipeline.
- Works with large datasets, databases and cloud platforms.
- Performs ETL (Extract, Transform, Load).
- Ensures data is clean, reliable and available for analysis.
- Uses tools like: SQL, Python, Azure, Databricks, Power BI (Power Query), etc.



Data Engineer

Output: Clean, transformed data in Data Warehouse / Data Lake.

Data Analyst

- Uses the prepared data to find insights.
- Creates reports, dashboards and visualizations.
- Works with tools like Power BI, Excel, SQL.
- Helps the business make data-driven decisions.
- Builds measures, calculated columns and DAX (in Power BI).

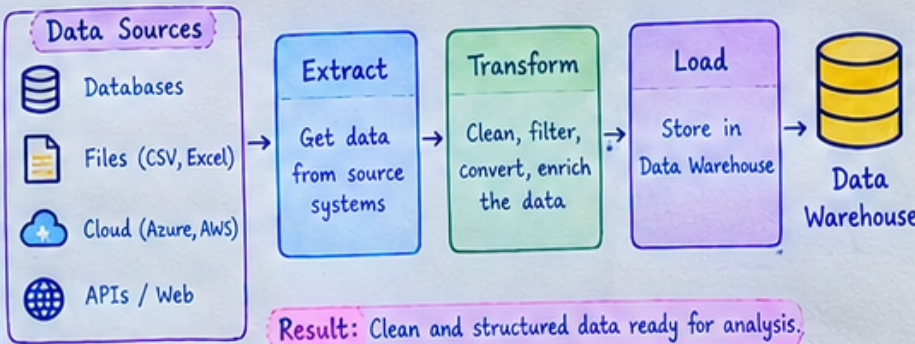


Data Analyst

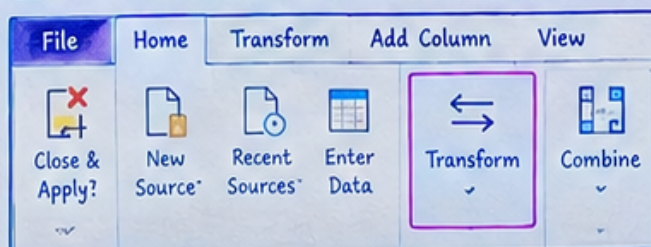
Output: Interactive reports and meaningful insights.

4. 1. ETL (Extract Transform Load)

The Data Engineer collects data from different source systems, extracts the data, transforms it (converts raw data into clean and usable data), stores it in a data warehouse and then loads it for analysis.



5. Power Query Editor - Transform the Data

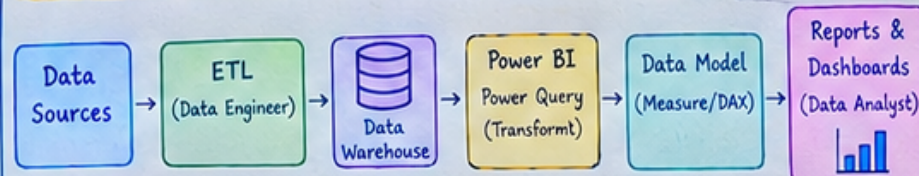


In Power BI, Power Query Editor is used to transform raw data into a proper and clean format.

Steps in Power Query:

- 1 Home - Connect to data source
- 2 Transform Data - Clean, filter, change data types, remove errors, etc.
- 3 Load - Load the data to Power BI model for analysis.

6. Overall Flow - From Data Source to Report



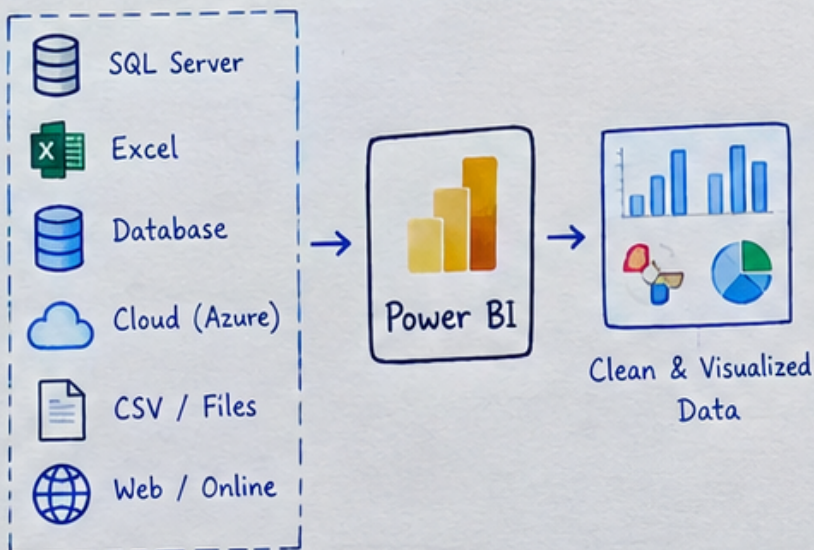
Summary:

- Data Engineer → Does ETL, builds data pipelines, manages data.
- Data Analyst → Uses clean data to create reports and insights.
- Power Query → Main tool for data transformation in Power BI.
- Front End → Reports, dashboards, visualizations (what users see).
- Back End → Data sources, Power Query, data model (what happens behind the scenes).

Types of Data Connector in Power BI

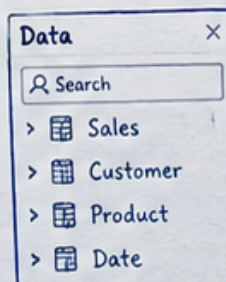
1. First Step

→ Extract the data from multiple data source system.

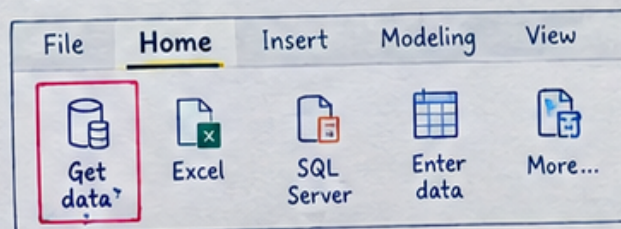


2. Collect the data using 3 options

1 Right side panel → Data (in the Power BI Desktop)



2 Home tab → Get data (in the ribbon)



3 Blank sheet → Add data to your report (in the report view)



Get Data - Types of Connectors

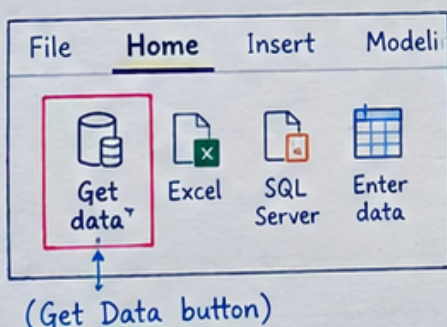
Power BI provides many connectors to get data from different sources.

- 1 File (Excel, CSV, XML, JSON, etc.)
- 2 Databases (SQL Server, MySQL, PostgreSQL, etc.)
- 3 Microsoft Fabric (Lakehouse, Warehouse, etc.)
- 4 Power platform (Dataverse, Power Apps, etc.)
- 5 Azure (Blob Storage, SQL DB, etc.)
- 6 Online services (SharePoint, Salesforce, etc.)
- 7 Other (OData, Web, etc.)

Example : Get Data → CSV → Company territory lookup.csv

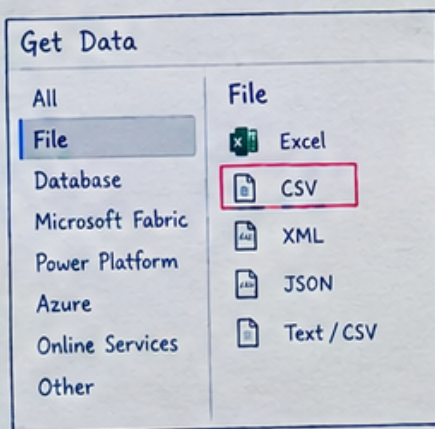
1 Click on Get Data

Go to Home tab → Click Get Data (or right side panel → Data).



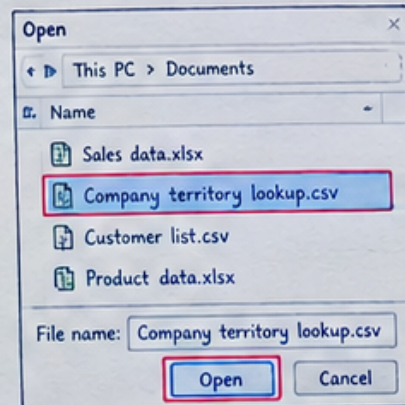
2 Choose CSV

In the Get Data window, select File → CSV.



3 Select the file

Choose the file Company territory lookup.csv from your system.



4 File origin & settings

- File origin : Windows / UTF-8 (default)
- Delimiter : Comma (,)
- Data type detection : ☒ (recommended)

→ Click Transform data → Power Query Editor is opened.

5 Power Query Editor

Here you can :

- Remove / change columns
- Change data types
- Fix errors
- Do transformations
- Then click Close & Apply to load the data to Power BI.

A screenshot of the Power Query Editor window. It shows a table with two columns: 'Territory' and 'Region'. The data is as follows:

Territory	Region
West	USA
East	USA
South	Canada
North	Canada

(Power Query Editor)

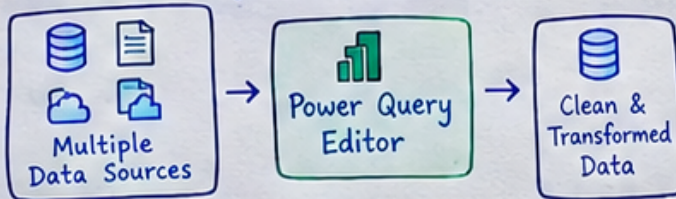
In short :

Data connectors help you bring data from different sources into Power BI. In this example, we used the File connector (CSV) to load a file, and then used Power Query Editor to clean and transform the data for better analysis and visuals.

Power Query Editor in Power BI

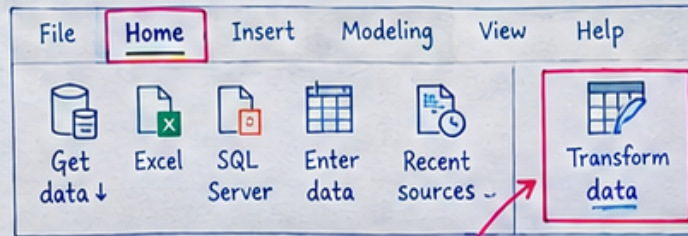
① What is Power Query Editor?

- Power Query Editor is a powerful tool in Power BI used to extract, transform and load (ETL) data from different sources.
- It helps to clean, shape and prepare data before loading it into your report.



② To open Power Query Editor

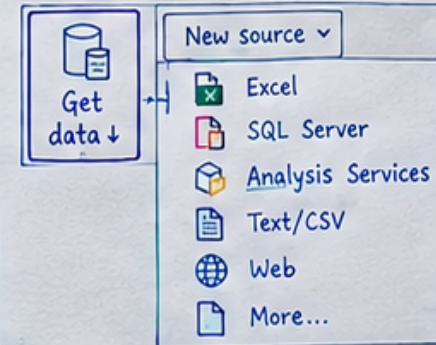
Home Tab → Transform data



Click on Transform data
→ Power Query Editor will open.

③ Data extraction

In the Home tab → New source →
You can extract data from multiple data sources.



Example:

- Excel file
- SQL Server
- Folder
- SharePoint
- Web
- etc.

④ For Transformation

In Power Query Editor, you can:
Example:

Change data type

Keep rows

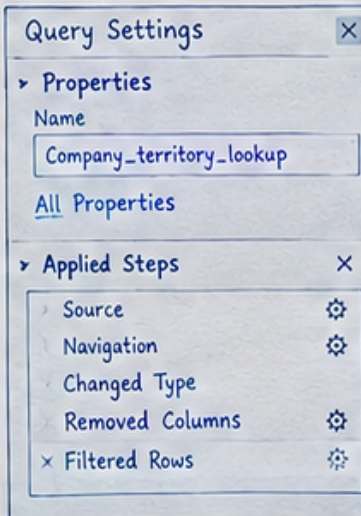
Remove columns

Remove rows

Name	Age
Kajal	25
Rohit	30
Sneha	28
Amit	32

- Change data type (e.g. Text → Number)
- Keep rows (e.g. Filter rows where Age > 25)
- Remove columns (e.g. remove unnecessary column)
- Remove rows (e.g. remove duplicates / blank rows)

⑤ Query Setting



Properties

- Name: You can change the name of the table.

Applied Steps

- Shows the list of changes or transformations applied to your data.

Example steps:

- Source (data source)
- Changed Type (data type)
- Removed Columns
- Filtered Rows
- Renamed Columns

⑥ Formula Bar

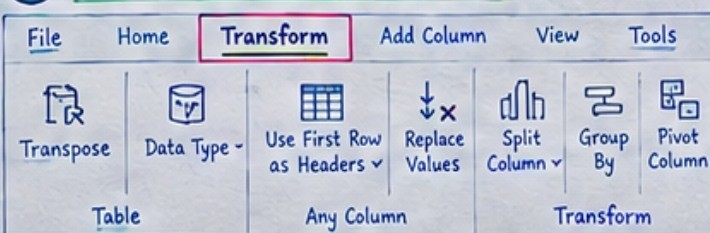
```
fx = Table.TransformColumnTypes(  
    #"Changed Type",  
    {"Territory", type text})
```

Shows the M code (Power Query language) behind your steps.

Where it is helpful?

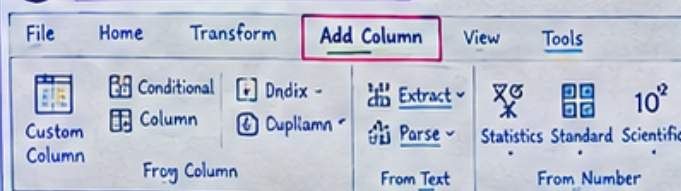
- View and edit the M code manually
- Understand the steps
- Reuse the code in other queries

⑦ Transform Ribbon (in detail)



- Transpose - switch rows and columns
- Data type - change column data type
- Use first row as headers - promote first row to headers
- Replace values - find and replace
- Split column - split text into multiple columns
- Group by - group data (like Pivot Table)
- Pivot column - unpivot or pivot data

⑧ Add Column (in detail)

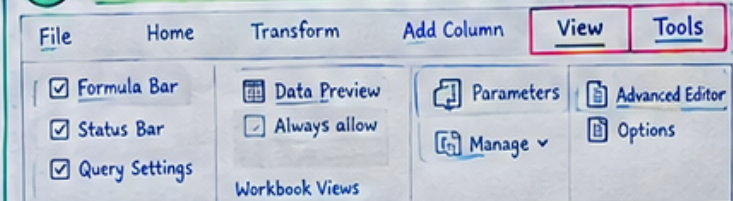


- Custom Column - create your own column using a formula
- Conditional Column - add column with conditions
- Index Column - add index
- Duplicate Column - copy existing column
- Extract - get part of text (e.g. first 3 letters)
- Parse - split text into parts (e.g. date, number)
- Statistics / Standard / Scientific - number functions

Example: Add a custom column

= Text.Upper([Name]) → to make name in uppercase

⑨ View and Tools Tab (in detail)



View tab

- Formula bar → show/hide
- Status bar - show/hide
- Query settings - open/close
- Data preview - preview data
- Always allow - faster loading

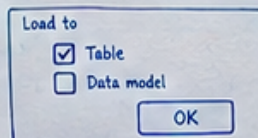
Tools tab

- Parameters - create parameters
- Manage parameters
- Advanced Editor - open M code
- Options - global settings

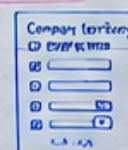
⑩ After Transformation - Load the Data

Click on Close & Apply (in Home tab)

Power Query Editor = Extract + Transform + Load
= Clean Data → Better Insights! 😊



Data will be loaded to
Power BI and you can
use it in your report.



Company	Territory	Region
ABC	West	USA
XYZ	East	USA
PQR	South	Canada

Tip:

Always check the applied steps and make sure the data looks correct before loading.

Basic Data Transformation

* In Power Query Editor, we can clean and shape the data using different transformation options.



1. Remove Columns

Used to delete unwanted columns from your data.

Before

ID	Name	Email	Phone	Country
1	Kajal	kajal@abc.com	12345	India
2	Rohit	rohit@xyz.com	67890	USA
3	Sneha	sneha@abc.com	45678	UK

Remove Phone column

After

ID	Name	Email	Country
1	Kajal	kajal@abc.com	India
2	Rohit	rohit@xyz.com	USA
3	Sneha	sneha@abc.com	UK

Path: Home tab → Remove Columns

2. Remove Rows

You can remove rows based on different conditions (top, alternate, duplicate, blank, errors).

	Remove Top Rows	→ removes first N rows
	Remove Alternate Rows	→ keeps every other row
	Remove Duplicates	→ removes duplicate rows
	Remove Blank Rows	→ removes empty rows
	Remove Errors	→ removes rows with errors

Example: Remove top 1 row

Name	Sales
Kajal	100
Rohit	200
Sneha	300

After

Name	Sales
Rohit	200
Sneha	300

Path: Home tab → Remove Rows → (option)

3. Change the name of table

- By default, the table name is based on the file name or source.
- You can give a meaningful name.

Query Settings

+ Properties

▼ Name

Sales_Data

Example:

Before: Table1

, After: Sales_Data

Path: Right side panel → Properties → Name

4. Change the name of column

Rename columns to make the data understandable.

Before

EmpID	EmpName	Dept
101	Kajal	IT
102	Rohit	HR
103	Sneha	Finance

After

Employee ID	Name	Department
101	Kajal	IT
102	Rohit	HR
103	Sneha	Finance

Path: Home tab → Transform → Use the column header to rename (or right click → Rename)

5. Check the data type

Helps to ensure the data is in the correct format (e.g. Text, Number, Date, etc.).

Column	Current Type	Correct Type
ID	Text	Whole number
Amount	Text	Decimal number
Date	Text	Date
IsActive	Text	Boolean

Path: Transform tab → Data type (icon)

Example: Change Amount from Text → Decimal number

6. Replace values

Used to replace specific values in a column.

Example: Replace "USA" with "United States"

Before

Country	Sales
USA	100
India	200
UK	300

After

Country	Sales
United States	100
India	200
UK	300

Path: Transform tab → Replace Values

7. Replace errors

Used to handle error values (e.g. #N/A, #DIV/0!, etc.).

Example: Replace errors with 0

Sales	Saler
100	100
#N/A	0
200	200
#DIV/0!	0
50	50

Path: Transform tab → Replace Errors

8. Split column

Used to split a column based on a delimiter (e.g. comma, space, etc.).

Example: Split Full Name into First Name & Last Name

Before	After
Full Name	First Name Last Name
Kajal Ubale	Kajal Ubale
Rohit Sharma	Rohit Sharma
Sneha Patil	Sneha Patil

Path: Transform tab → Split Column → By Delimiter (or By Positions)

9. Column - Change Type - Rename the applied step

After changing the data type of a column, the applied step will be shown as a new name (e.g. Changed Type).

Steps

Name	Applied Step
Source	Source
▼ Changed Type	Changed Type
Renamed Columns	Renamed Columns
Filtered Rows	Filtered Rows

↑ If you change the data type again, it will appear as a new step (e.g. Changed Type1).

Example:

- Column: Amount
- From: Text
- To: Decimal Number
- Applied Step Name: Changed Type

Final Result:

ID	Amount
1	100.5
2	200.75
3	300.00

Tip:

- Always check the applied steps to know what changes are made.
- You can reorder, delete or rename the steps if needed.
- Use the Preview feature to check the data before loading.

Clean Data → Better Analysis → Better Insights 😊

Assignment No : 1 (Power BI - Query Editor)

Task : Work with two new CSV files and perform data transformation steps.

1 Create queries to connect to the two new .csv files

- Go to Home tab → Get data → Text/CSV.
- Select the first file (product_category_lookup.csv) → Load.
- Repeat for second file (product_subcategory_lookup.csv) → Load.

Get Data



File



Database



Text/CSV



Web



More...

Result :

Two queries will be created in Power Query Editor.

2 Name your queries

- Right click on each query → Rename.

Queries [2]

Product Category Lookup

Product Subcategory Lookup

Example :

Query 1: Product Category Lookup

Query 2: Product Subcategory Lookup

3 Confirm column headers & data types

- Check that column headers are promoted (Use the first row as headers).
- Verify data types (Text, Whole number, etc.).

Product Category Lookup		
CategoryID	CategoryName	Status
1	Beverages	Active
2	Snacks	Active
3	Household	Active

Data type check

- CategoryID → Whole number
- CategoryName → Text
- Status → Text

In Power Query Editor → Home tab → Use "Promote Headers" (if needed) and check Data Type icon.

4 Add a new column to extract all characters before the dash ("-") in the product SKU column and name it "SKU Type".

Use the Add Column tab → Custom Column.

Formula:

= Text.BeforeDelimiter ([Product SKU], "-")

Example :

Before Transformation

Product SKU	SKU Type (new column)
ABC-123-RED	ABC
DEF-456-BLUE	DEF
GHI-789-GRN	GHI

5 Update the SKU Type calculation to return all characters before the second dash, instead of the first.

Use the same Custom Column option.

Formula:

= Text.BeforeDelimiter (Text.AfterDelimiter ([Product SKU], "-"), "-")

After Update

Product SKU	SKU Type
ABC-123-RED	123
DEF-456-BLUE	456
GHI-789-GRN	789

- (Now it returns the value between first and second dash)

6 Replace zeros(0) in the product style column with "NA".

Use Transform tab → Replace Values.

Find what: 0 Replace with: NA

Example :

Product Style (Before)	Product Style (After)
0	NA
StyleA	StyleA
0	NA
StyleB	StyleB

7 Close and load to your data model

- Click Close & Apply (Home tab).
- The transformed data will be loaded into Power BI data model.



Close & Apply

Navigator

- ☒ Product Category Lookup
- ☒ Product Subcategory Lookup

Result :

Both queries will be loaded to the data model and ready for use in reports and visuals.

Final Output (Example - Product Category Lookup)

CategoryID	CategoryName	Status
1	Beverages	Active
2	Snacks	Active
3	Household	Active

Final Output (Example - Product Subcategory Lookup)

SubcategoryID	CategoryID	SubcategoryName	SKU Type	Product Style
101	1	Soft Drinks	123	NA
102	1	Juices	456	StyleA
201	2	Chips	789	NA

Key Points to Remember :

- ✓ Use Text/CSV to connect .csv files.
- ✓ Rename queries for easy identification.
- ✓ Promote headers & check data types.
- ✓ Use Custom Column for new calculations.
- ✓ Use Text.BeforeDelimiter / Text.AfterDelimiter.
- ✓ Replace zeros with "NA".
- ✓ Click Close & Apply to load data.



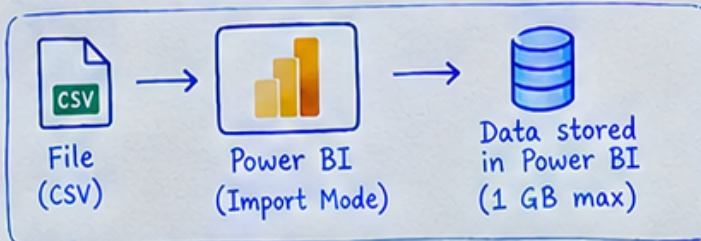
Storage and Connection Mode

Power BI

Different modes to connect and store your data in Power BI.

1. Import Mode

Works on exported data (like CSV file).



- Reserves the memory of Power BI.
- Can store upto 1 GB (per dataset).
- Data should not be changed.
- Data remains same as it is.

Where it is used ?

- When data is small to medium size.
- When you need faster performance (no dependency on source).
- When data is not changing frequently.

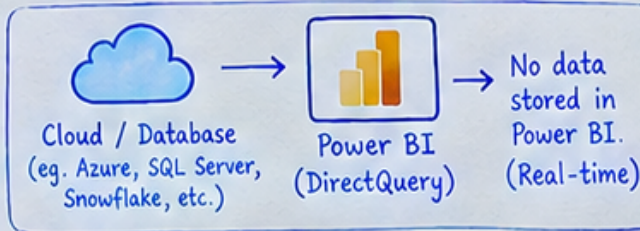
Example :

- Load a CSV file of sales data into Power BI using Import mode.



2. DirectQuery Mode

Accesses data from cloud storage / main source directly.



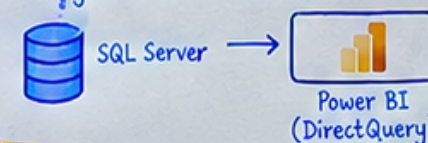
- Make sure that data can be accessed from the main source only.
- Source data is getting changed frequently.

Where it is used ?

- When data is large and cannot fit in memory.
- When you need real-time / up-to-date data.
- When data changes frequently.

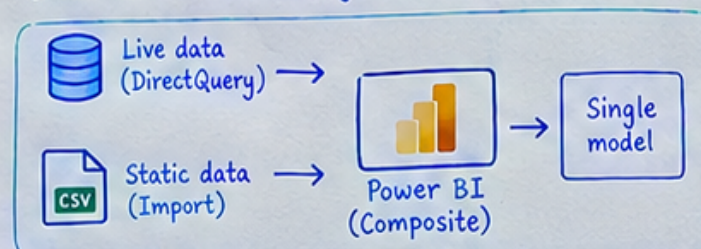
Example :

- Connect to a SQL Server / Azure database using DirectQuery mode to get live sales data.



3. Composite Mode

Uses both DirectQuery and Import mode together.



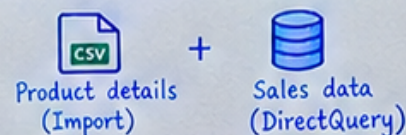
- Combines DirectQuery and Import in one model.
- Works with a single dataset and single model.
- Gives flexibility to use both real-time and historical data.

Where it is used ?

- When you have both large real-time data and small static data.
- When you want to analyze data from different sources in one report.

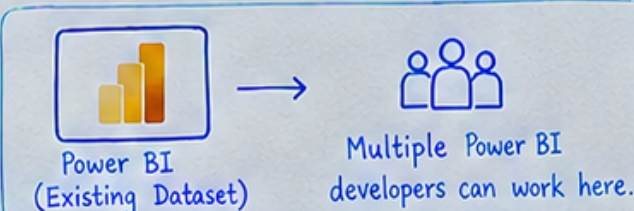
Example :

- Import static product details (CSV) and use DirectQuery for live sales data in one model.



4. Live Connection

Connects directly to an existing Power BI dataset.



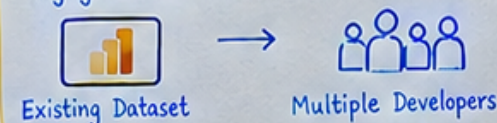
- No data is stored in your report.
- Connects to another Power BI dataset.
- Multiple developers can work on the same dataset and report.

Where it is used ?

- When you want to reuse data and calculations from an existing Power BI dataset.
- When multiple developers need to work on the same dataset.

Example :

- Connect to an existing Power BI dataset shared by your team.



Quick Comparison

Mode	Data Storage	Data Source	Data Update	Best For
Import	Stored in Power BI (upto 1GB)	File (CSV, Excel, etc.)	Manual refresh	Small/medium data, better performance
DirectQuery	Not stored (live connection)	Cloud / Database	Real-time	Large data, live data, frequent changes
Composite	Both (Import + DirectQuery)	Multiple sources	Depends on source	Mix of real-time + static data
Live Connection	Not stored (uses existing dataset)	Existing Power BI dataset	Real-time	Collaboration, reuse of existing data

Key Takeaway :

- ✓ Import = fast & simple (small data).
- ✓ DirectQuery = real-time (large data).
- ✓ Composite = best of both worlds.
- ✓ Live Connection = collaboration & reuse.

Choose the mode based on your data size, update frequency and business needs. 😊

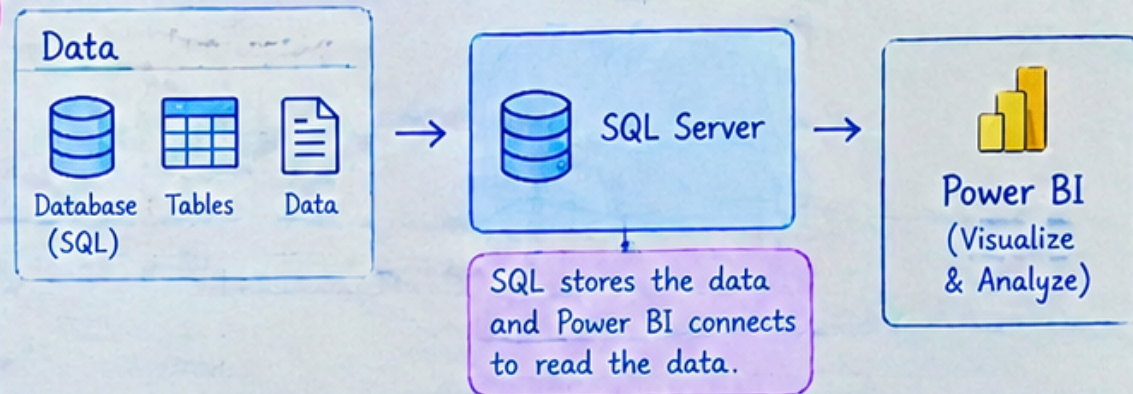
Connecting to SQL Database

SQL Database stores data in structured tables and helps us manage, query and analyze the data efficiently.

1. Why connect to SQL database? (Where it is used?)

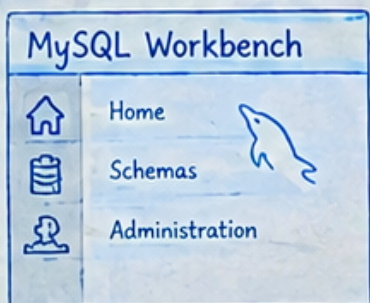
- Centralized data storage – Keeps data in one place.
- Handles large amount of data efficiently.
- Supports complex queries and joins.
- Used in real-time reporting and dashboards.
- Helps in data analysis, business intelligence and decision making.

Used in: Reporting, Analysis, Dashboards, Data warehousing, Business applications, etc.



2. Steps in SQL Workbench

1 Open MySQL Workbench



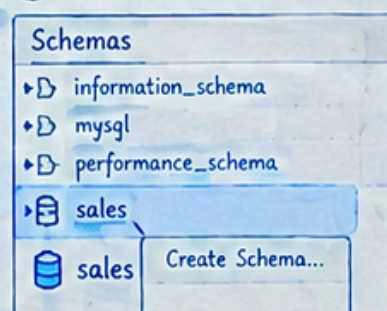
Open MySQL Workbench and click on "+" to create a new connection.

2 Create a new connection



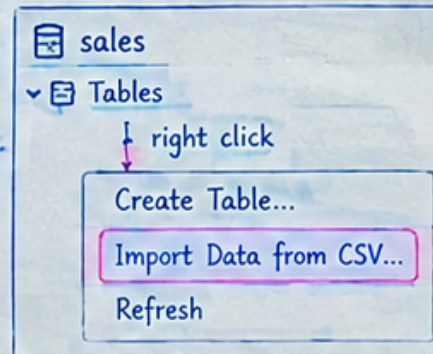
Give a name, enter server name, port, username and click "Test Connection" → OK.

3 Create a database



Right click on Schemas → Create Schema. (e.g. sales)

4 Go to table → Right click → Import data from CSV



Example:

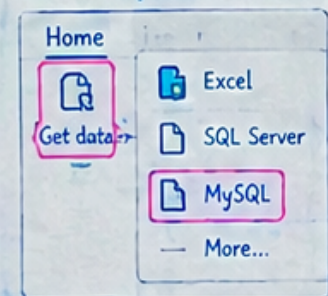
- After import, the CSV data will be loaded into a table in the sales database.

sales.customer

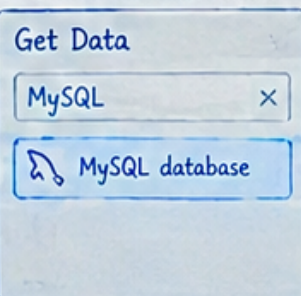
ID	Name	Email
1	Kajal	kajal@abc.com
2	Rohit	rohit@xyz.com
3	Sneha	sneha@abc.com

3. Import the data from MySQL to Power BI

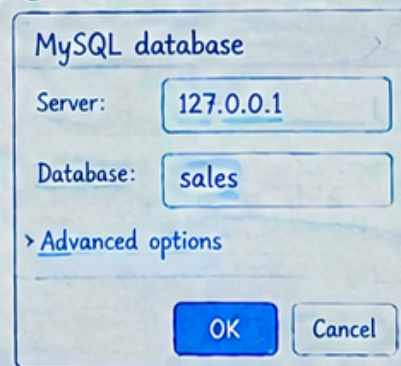
1 Open Power BI Desktop → Get data



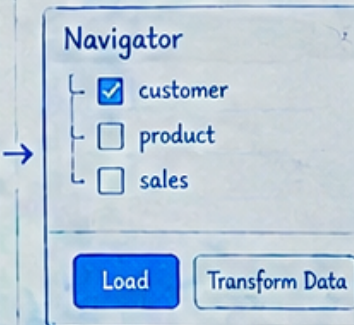
2 Choose MySQL connection



3 Enter server & database



4 Select tables → Load or Transform data



4. Do we need any connector?

- Yes. For MySQL, you need the MySQL Connector/NET (Oracle).
- Install the MySQL Connector/NET on your machine.
- After installation, restart Power BI.
- Then you can connect using the MySQL connector.

💡 If you don't install the connector, you may get an error like "DataSource.MissingClientLibrary",

★ Summary



Key Takeaway?

- ✓ Install MySQL Connector/NET.
- ✓ Use MySQL connector in Power BI.
- ✓ For MySQL, Import mode is commonly supported (DirectQuery may not be available).
- ✓ Always check the connector requirement before connecting. 😊

Website Data Gathering



Website data gathering means collecting data from websites (public web pages) and bringing it into Power BI for analysis and visualization.

① Why we use website data gathering?

- To get the latest and real-time information from websites.
- Useful when data is not available in a database or file.
- Helps to build dynamic reports (data can change on the website).
- Saves time – no manual data entry.

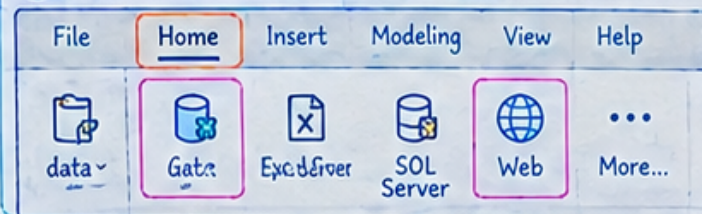
💡 In order to have dynamic data, we can gather the information using web scraping option.

② How to get website data in Power BI?

① Open Power BI Desktop



② Go to Home tab → Get Data



③ Select Web → Paste the website URL and click OK.

④ Power BI will fetch the data and you can transform it in Power Query Editor.

③ Example

- Let's say we want to get data from a website that shows latest stock prices.

Website URL (example):

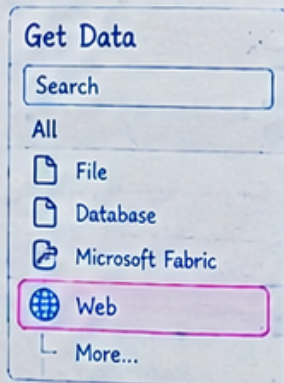
<https://www.nseindia.com/api/market-data/live-equity-market>

🌐 NSE - Live Market Data

Symbol	LTP	Change	%Change
RELIANCE	2,987.45	+12.30	+0.41%
TCS	3,456.10	+18.20	+0.53%
INFY	1,542.60	-5.10	-0.33%

- We can copy this URL and use web connector in Power BI to get this data.

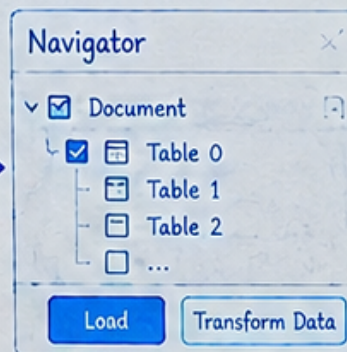
④ Visual Steps in Power BI



1. Select Web from Get Data.



2. Paste the website URL and click OK.



3. Select the table and click Load or Transform Data.

Result in Power BI

Symbol	LTP	Change	%Change
RELIANCE	2987.45	12.30	0.41%
TCS	3456.10	18.20	0.53%
INFY	1542.60	-5.10	-0.33%

4. Data is loaded into Power BI.

⑤ No need to use Python library

- Power BI has a built-in Web connector to get data from websites.
- You don't need to use Python or any external libraries for basic web data gathering.
- Python is only required when you need advanced web scraping or complex custom logic.



For most websites, the Web connector in Power BI is enough.

⑥ Where is it used?

- 📊 Market & Finance – stock prices, currency rates, financial news.
- ☁️ Weather Data – live weather, temperature, forecasts.
- 📰 News & Blogs – latest headlines, articles.
- 🛒 E-commerce – product prices, reviews, stock availability.

- ✈️ Travel & Hospitality – flight prices, hotel data.
- ⚽ Sports – live scores, match results.
- 🏛️ Government – public data portals.
- 🌐 Any public website – data that is available on the web.

★ Key Takeaway?

Website data gathering helps us bring live and dynamic data from websites into Power BI without using any additional libraries or coding.

